

Superstore Sales Data: Data Analysis & BI Reporting

OVERVIEW

This project leverages **IBM Cognos Analytics** to transform raw data into actionable business intelligence. By utilizing advanced reporting and interactive dashboards, the project identifies key trends, optimizes data exploration, and provides a clear narrative for stakeholders.

OBJECTIVES

- **Data Modeling** : Create a robust data module to ensure accurate reporting.
- **Interactive Visualization** : Design intuitive reports for data exploration.
- **Business Insights** : Extract high-impact findings to support data-driven decision-making.
- **Technical Mastery** : Demonstrate proficiency in IBM Cognos tools, including calculations, filters, and conditional formatting.

DATASET PROFILE

Name : Superstore Sales Data

Source : <https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting?resource=download>

Format : CSV

Dimensions :

```
PS C:\Users\tabde\Desktop\MCA-DS-2nd Sem\Descriptive Analysis> python.exe .\main.py
Number of rows: 9800
Number of columns: 18
```

Domain : Sales

Description : The core focus of the analysis is to provide a report style interface to users to provide a navigation of sales by region using prompting.

Name : Rishiraj Singh

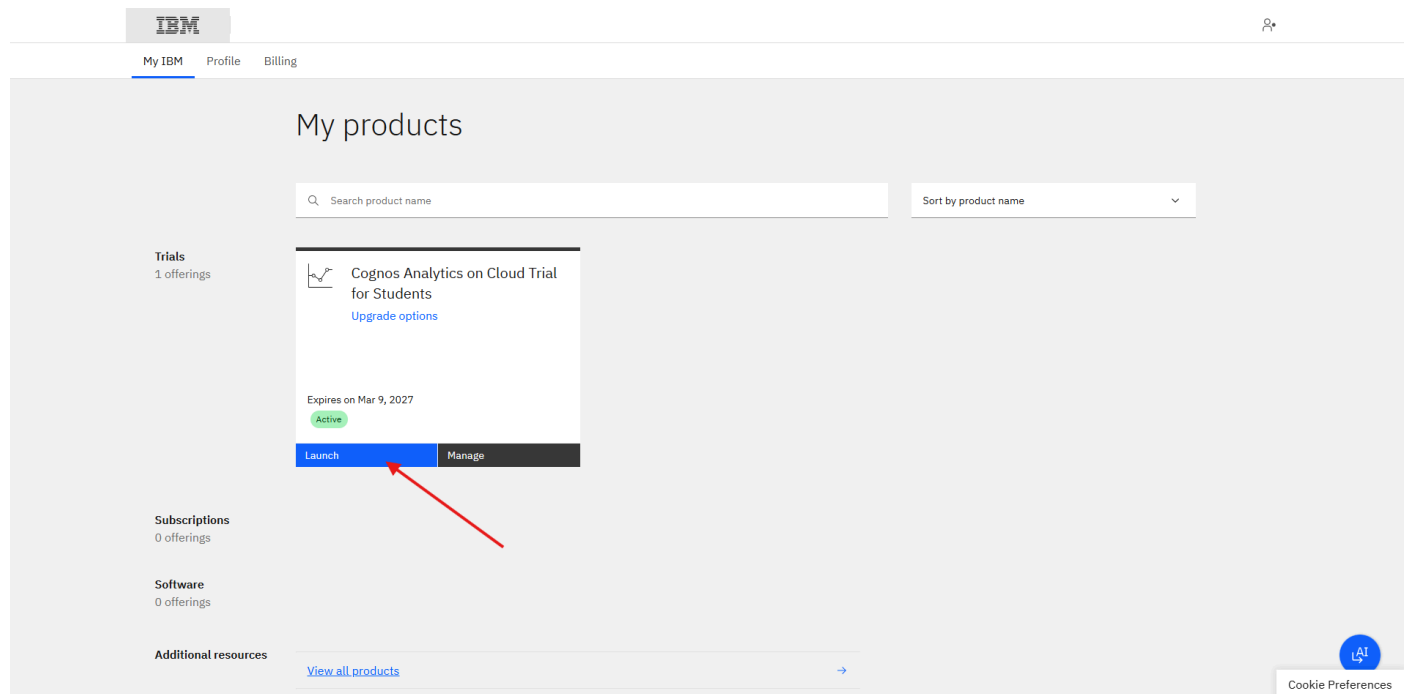
Course : Masters of Computer Applications in Data Science and Artificial Intelligence

Subject : Descriptive Analytics

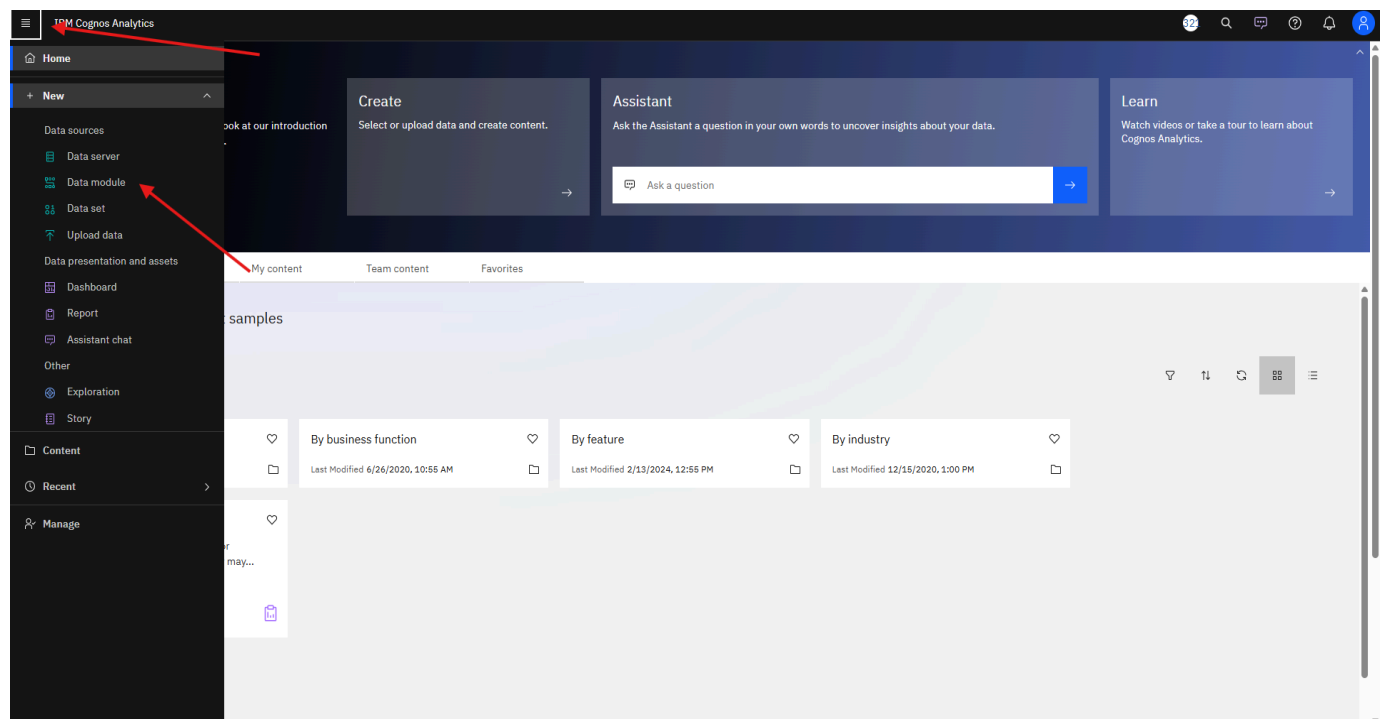
Roll Number : **1250259044**

Section : MCADS11

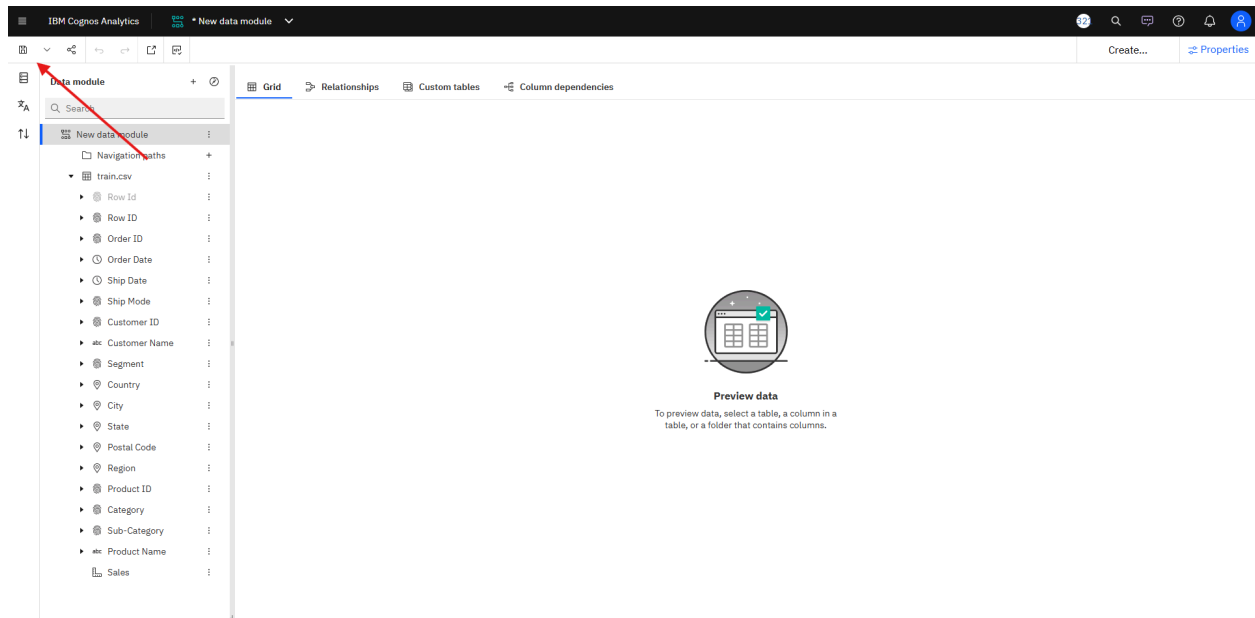
Step 1 : Login to Your IBM Cognos Account and launch IBM Cognos



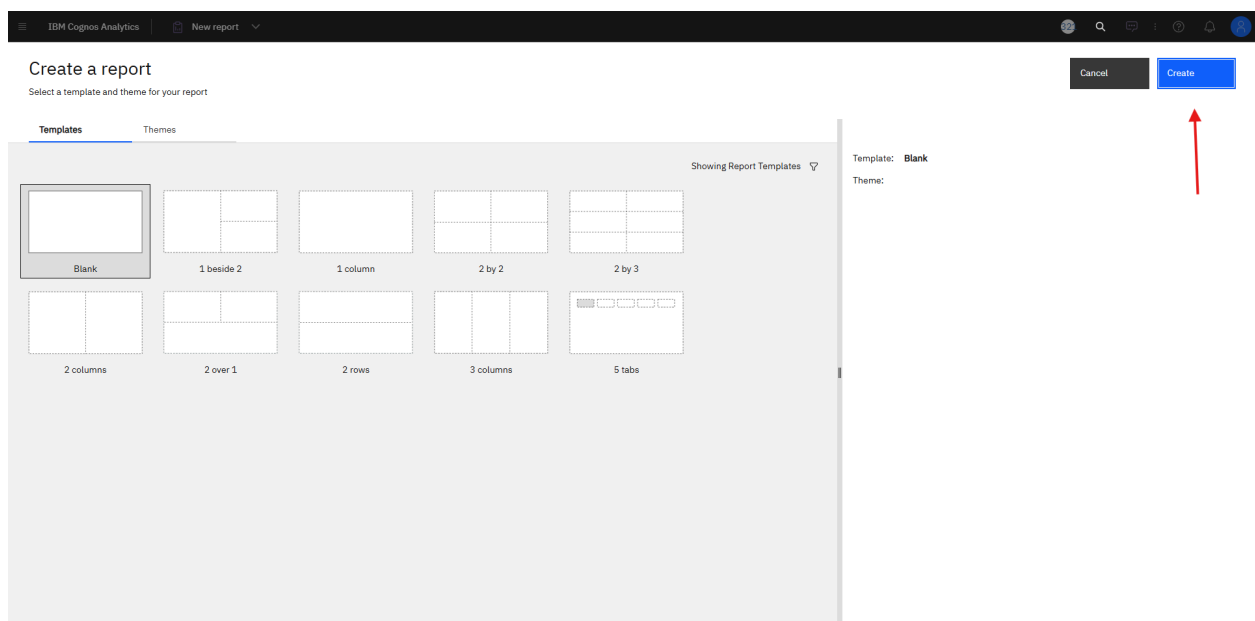
Step 2 : Create a new data module and load your data module to use in a report



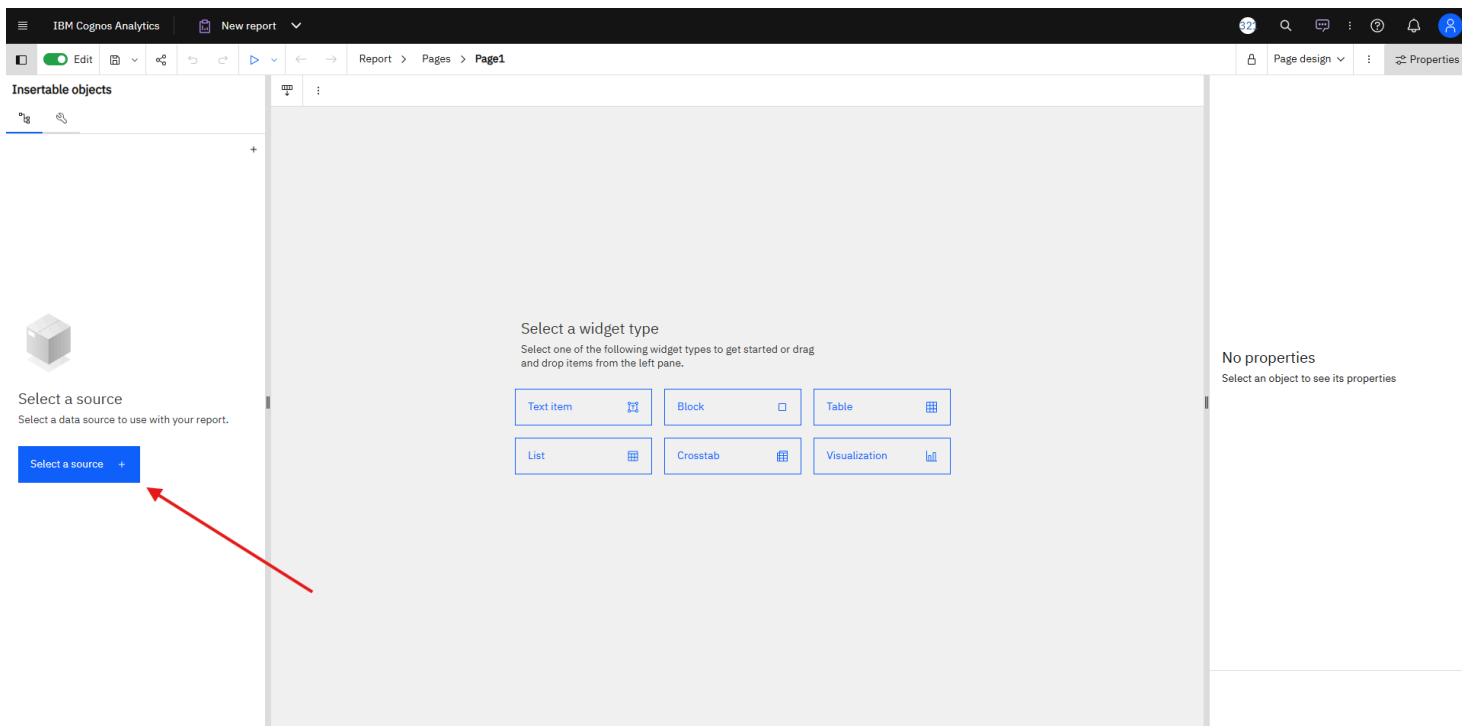
Step 3 : Save your data module and rename it, make sure to save it into **My Contents**.



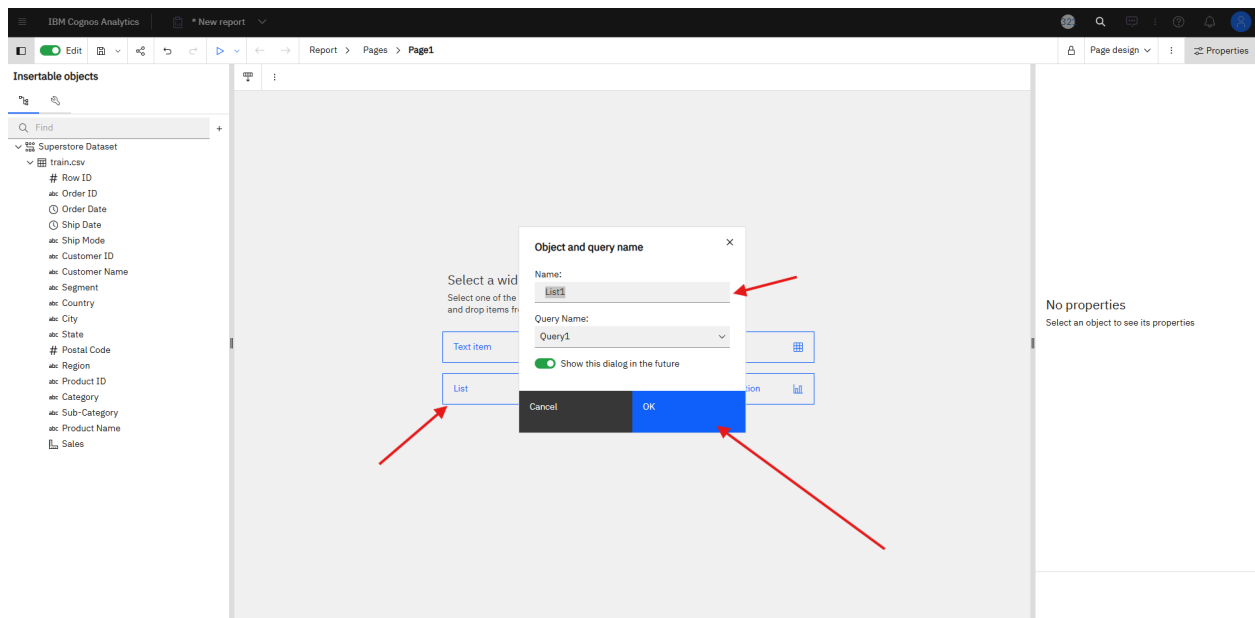
Step 4 : Navigate towards hamburger menu and Create a New Report with blank template.



Step 5 : Select Datasource and import your dataset under my contents to explore the data.



Step 6 : Create a list container and name it



Step 7 : Import the columns you need and run it as html to view your report

IBM Cognos Analytics - New report

Report > Pages > Page1

Insertable objects

- Superstore Dataset
 - train.csv
 - Row ID
 - Order ID
 - Order Date
 - Ship Date
 - Ship Mode
 - Customer ID
 - Customer Name
 - Segment
 - Country
 - City
 - State
 - Postal Code
 - Region
 - Product ID
 - Category
 - Sub-Category
 - Product Name
 - Sales

Drop items here to create new columns

Page design | Properties

Find

- CONDITIONAL
 - Conditional styles
 - Style variable
 - Render variable
 - No data contents
- DATA
 - Grouping & sorting
 - Query
 - Query1
 - Rows per page
 - Master detail relationships
 - Suppression
 - Properties
 - Share result set
- GENERAL
 - Contents height
 - Column titles
 - Scrollable area height
 - Pagination
 - Render page when empty
 - Highlight groups
- INTERACTIVE BEHAVIOR
 - Set parameter values
 - Container highlight
- BOX
 - Border
 - Margin
 - Box type
- COLOR & BACKGROUND
 - Background image

IBM Cognos Analytics - New report

Report > Pages > Page1

Insertable objects

- Superstore Dataset
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 - Row ID
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 - Postal Code
 - Region
 - Product ID
 - Category
 - Sub-Category
 - Product Name
 - Sales

Row ID Ship Date Order Date Sales Region

<Row ID> <Ship Date> <Order Date> <Sales> <Region>

<Row ID> <Ship Date> <Order Date> <Sales> <Region>

<Row ID> <Ship Date> <Order Date> <Sales> <Region>

Page design | Properties

Find

- CONDITIONAL
 - Conditional styles
 - Style variable
- BOX
 - Border
 - Padding
 - Box type
- COLOR & BACKGROUND
 - Background image
 - Background effects
 - Background color
 - Foreground color
- FONT & TEXT
 - Font
 - Horizontal alignment
 - Vertical alignment
 - White space
 - Spacing & breaking
 - Direction & justification
- POSITIONING
 - Size & overflow
- MISCELLANEOUS
 - Classes

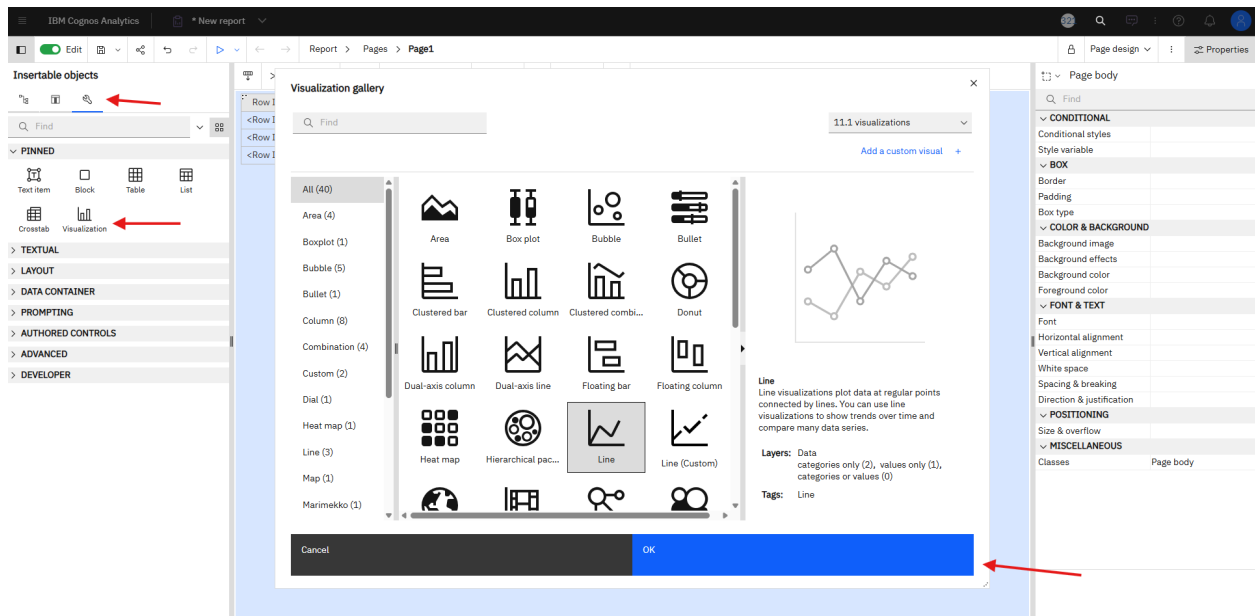
Maintenance: The upgrade is now complete. Click on More Info to see what actions may be necessary and to subscribe to future events.

IBM Cognos Analytics - New report

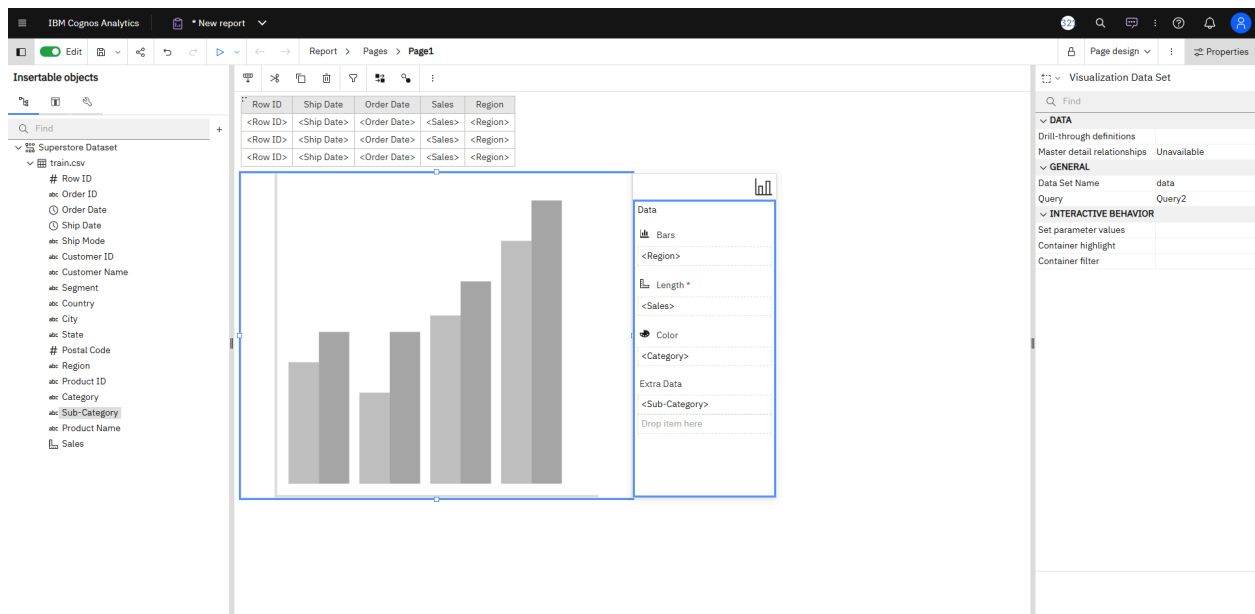
Mode: Interactive View | Filters

Row ID	Ship Date	Order Date	Sales	Region
2	Nov 11, 2017	Aug 11, 2017	731.94	South
4	Oct 18, 2016	Nov 10, 2016	957.5775	South
7	Jun 14, 2015	Sep 6, 2015	7.28	West
11	Jun 14, 2015	Sep 6, 2015	1,706.184	West
14	Oct 12, 2017	May 12, 2017	407.976	West
16	Nov 26, 2016	Nov 22, 2016	2.544	Central
20	Sep 1, 2015	Aug 27, 2015	213.48	West
28	Sep 21, 2016	Sep 17, 2016	3,083.43	East
29	Sep 21, 2016	Sep 17, 2016	9.618	East
31	Sep 21, 2016	Sep 17, 2016	3.264	East
35	Oct 23, 2018	Oct 19, 2018	29.472	Central
37	Oct 12, 2017	Dec 8, 2017	190.92	Central
43	Jul 22, 2017	Jul 17, 2017	77.88	West
44	Sep 23, 2018	Sep 19, 2018	95.616	South
52	Apr 22, 2016	Apr 18, 2016	6.16	Central
56	Jun 18, 2017	Jun 17, 2017	208.56	East
58	Jun 18, 2017	Jun 17, 2017	319.41	East
66	Nov 30, 2016	Nov 24, 2016	79.76	West
74	May 2, 2016	Apr 26, 2016	97.04	South
76	Dec 11, 2018	Dec 9, 2018	1.248	Central

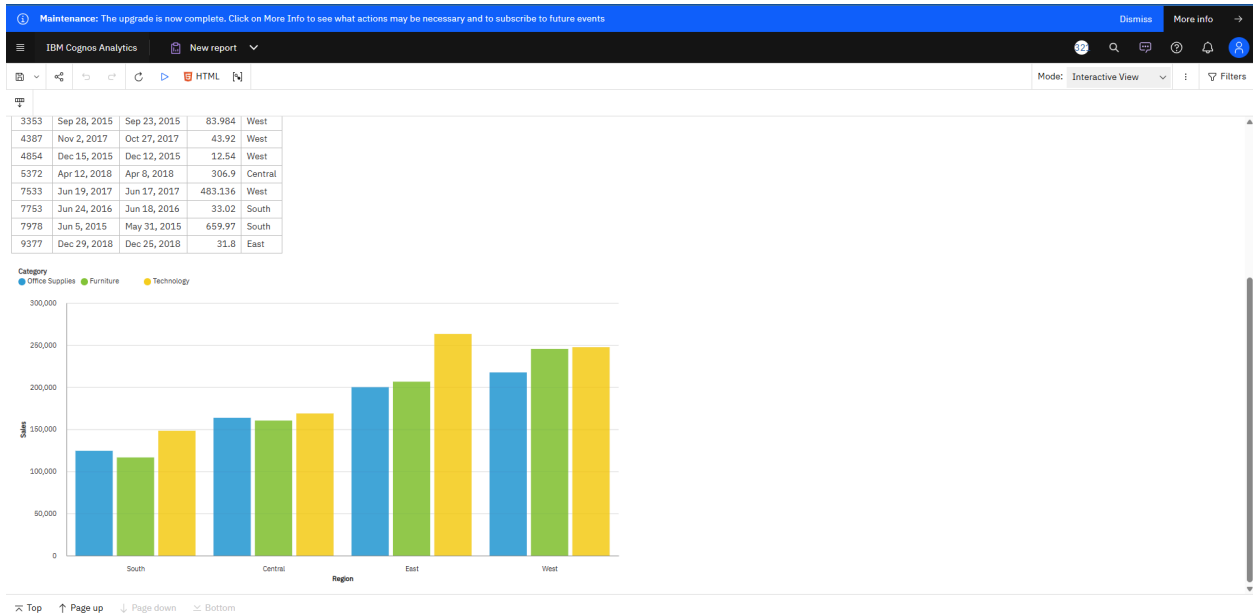
Step 8 : Go back to your report tab, within **insertable objects** there is a tool icon, click on the icon and insert the visualization chart you want.



Step 9 : Go back to the sources and drag the x and y axis according to your dataset.



Step 10 : Run the report for intuitive visualizations with data



Step 11 : Create a custom filter to explore data of your interest by clicking on filter

IBM Cognos Analytics New report

Report > Pages > Page1

Insertable objects

Find

Superstore Dataset

train.csv

- # Row ID
- # Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer ID
- Customer Name
- Segment
- Country
- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales

Filter condition - Ship Date

Range Individual

Add a condition

Keep these values

Between

Aug 1, 2017

Dec 31, 2018

Advanced settings

Cancel OK

List column title

Find

CONDITIONAL

Conditional styles

Style variable

Text source variable

TEXT SOURCE

Source type

Data item label

DATA

Data format

Drill-through definitions

BOX

Border

Padding

Box type

COLOR & BACKGROUND

Background image

Background effects

Background color

Foreground color

FONT & TEXT

Font

Horizontal alignment

Vertical alignment

White space

Spacing & breaking

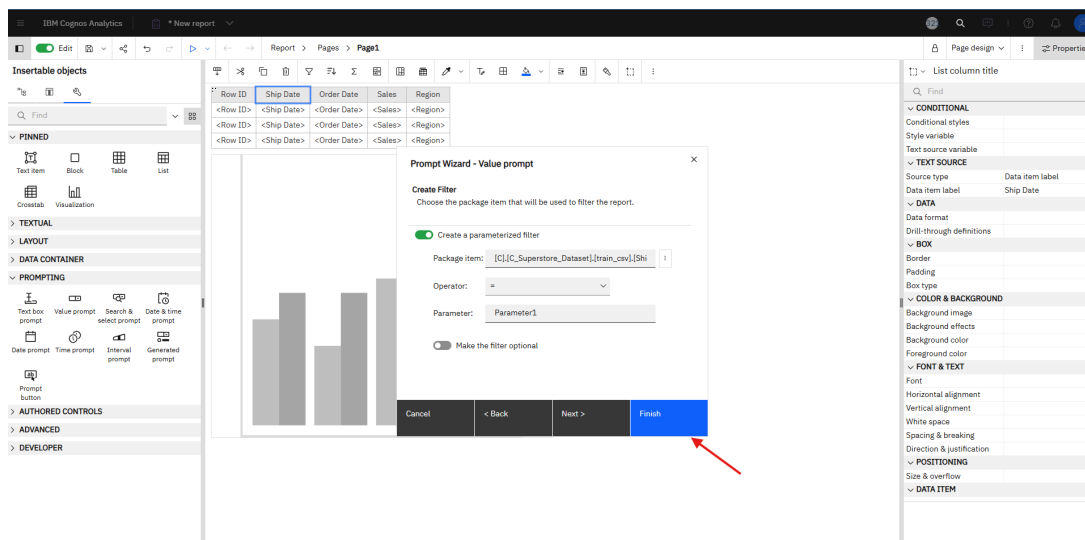
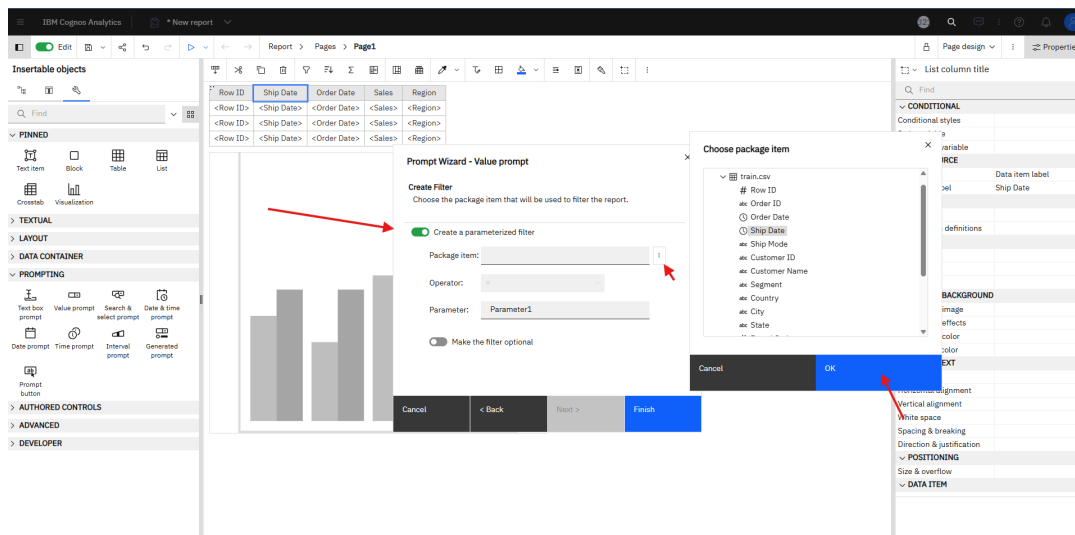
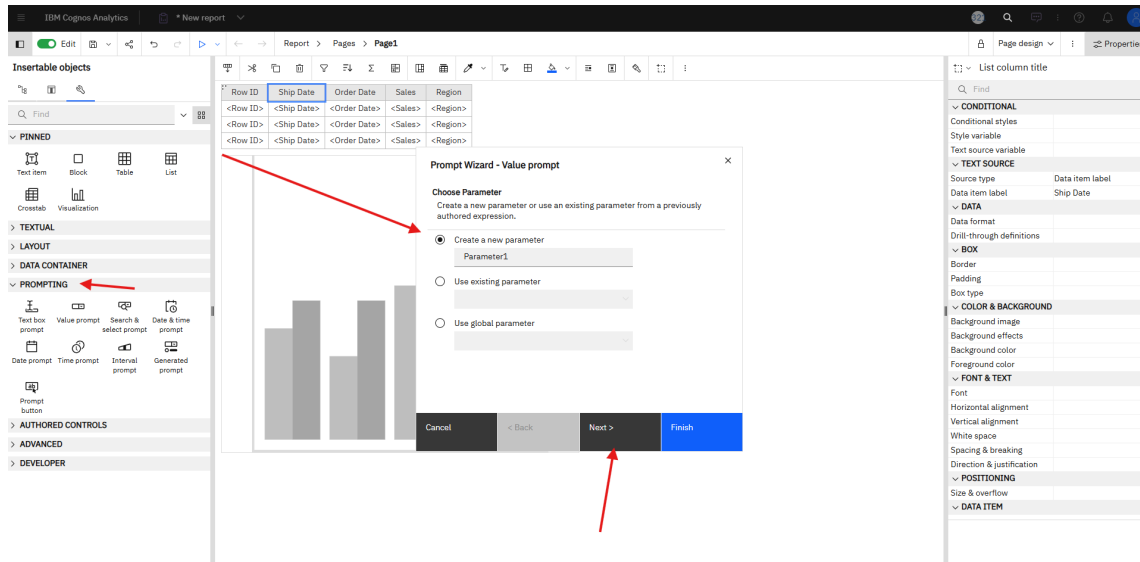
Direction & justification

POSITIONING

Size & overflow

DATA ITEM

Step 12 : Making the report interactive using prompts, unfold the prompting menu and select the type prompt you want and create a parameter for the prompt.



Step 13 : To create a **drill through summary**, on the right panel look for drill through definitions and add an item to the list.

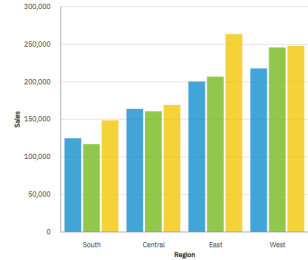
Step 14 : Run the final report

up

Row ID	Ship Date	Order Date	Sales	Region
1846	Jan 17, 2018	Jan 12, 2018	40.08	East
1847	Jan 17, 2018	Jan 12, 2018	37.68	East

Category

● Office Supplies ● Furniture ● Technology



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